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Paleomagnetism and Cyclostratigraphy of the Permian-Triassic Boundary Interval of the Staroe Slukino Section, Vladimir Region

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Abstract—The paper presents the results of paleomagnetic, rock magnetic, and cyclostratigraphic studies of continental red beds that compose the Permian-Triassic boundary interval of the Staroe Slukino section in the Vladimir Region, Russia. Based on the directions of the characteristic components of the natural remanent magnetization of the studied rocks in the sampled stratum, the intervals of normal and reversed polarity related to the regional magnetozones r_2RnP, r_3RnP, and N_3P-T were identified. Within the r_3RnP zone, there is an interval of anomalous paleomagnetic directions, which has similar characteristics to those identified earlier in the coeval intervals of the Nedubrovo, Zhukov Ravine, and Okskiy Siyezd sections. Revising the biostratigraphy allows us to assume that the two zones of anomalous paleomagnetic directions in the composite magnetic polarity scale of the upper Permian of the Russian Platform are a reflection of the same epoch of an anomalous geomagnetic field configuration. It is estimated by the cyclostratigraphic method that the studied 16 m interval of the Staroe Slukino section took 900 ± 20 thousand years to accumulate, which limits the duration of the geomagnetic field anomalous state near the Permian-Triassic boundary to ~110 thousand years. A new Permian-Triassic (~252 Ma) paleomagnetic pole of the East European Platform is calculated: plat = 36.3°; plong = 155.0°; dp/dm = 2.8°/4.8°.

Keywords: paleomagnetism, East European Platform, Russian Platform, Permian, Triassic, magnetostratigraphy, cyclostratigraphy, paleomagnetic pole

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INTRODUCTION

The stratigraphic correlation of continental sedimentary rocks is often a difficult problem that requires a comprehensive approach to be solved reliably, including the detailed study of each particular rock outcrop. The absence of areal sedimentation and its discontinuous nature, numerous scour marks and deep, up to 10–15 m, alluvial trenches—all these complicate the recognition and correlation of coeval strata not only in sections that are distant from each other, but even in those that are close to each other. Lithological and biostratigraphic methods widely used for stratigraphic correlation of terrigenous strata does not always yield unambiguous results, so they are often used together with the techniques of paleomagnetology—magnetostratigraphy and rock magnetism. The paleomagnetic method has proved to be quite effective in solving a number of stratigraphic problems, such as (1) a detailed subdivision of rock sequence based on their paleo- and rock magnetic properties, (2) an identification of local and regional marker beds in the stratigraphic sequence, (3) a correlation of regional and local stratigraphic units with the Global Stratigraphic Scale (Molostovsky and Khramov, 1997).

When rocks of continental genesis are studied, stratigraphically complete sections with good paleomagnetic properties are hard to come by; therefore, the reliability and completeness of the developed regional magnetostratigraphic scales largely depend not only on the quality but also on the quantity of the investigated sections. In recent years, our team has been conducting a comprehensive study of sedimentary rocks of continental origin in the Permian-Triassic boundary interval in different regions of the Russian Basin of the East European Platform (Fetisova et al., 2018a; 2018b; 2020; 2022). The main objectives of the studies are to obtain reliable paleomagnetic properties and to correlate sections in order to improve and specify the regional magnetostratigraphic chart of the Permian-Triassic boundary beds of the East European Platform, to compare it with synchronous marine and continental sections of Western Europe and Asia, to use paleomagnetic data to estimate the main characteristics of the Earth's magnetic field at the

Paleozoic-Mesozoic boundary, and to improve paleotectonic reconstructions of Eurasia around 250 Ma.

Finding intervals of the anomalous configuration of the Earth's magnetic field in the geological record is important because it allows us not only to study its evolution, but also to obtain a reliable magnetostratigraphic benchmark, which can be used for the purposes of global stratigraphic correlation. Stable anomalous paleomagnetic directions have been detected in rocks of different age and origin (Khramov, 2007; Khramov and Iosifidi, 2012; Shatsillo and Pavlov, 2019), including the Permian-Triassic boundary rocks of the Russian Platform—the Nedubrovo section of the Vologda Region (Fetisova et al., 2018). Recently, based on the results of comprehensive biostratigraphic and paleomagnetic studies of the Permian-Triassic strata in sections of the Central Russian Basin—Zhukov Ravine (Vladimir Region), Slukino (Vladimir Region), and Okskiy Siyezd (Nizhny Novgorod Region), we (Fetisova et al., 2022) have hypothesized the existence of two closely spaced intervals of the anomalous configuration of the geomagnetic field in the Late Permian and shown a more complex structure of the magnetic polarity scale for the upper Permian interval than that suggested in (Hounslow and Balabanov, 2018). Nevertheless, it is necessary to obtain new paleomagnetic data in order to prove the existence of the regional magnetic zone r_3RnP identified in (Fetisova et al., 2022), as well as two anomalous paleomagnetic zones a1r4 and a2r4 in the magnetostratigraphic sequence of the Oka River basin. In this paper, we present the results of the paleomagnetic studies of the Upper Permian-Lower Triassic sedimentary rocks in the Staroe Slukino section (Vladimir Region), which were carried out in 2021–2022, and their interpretation. In addition, we have estimated the duration of the studied rocks accumulation based on the cyclostratigraphy analysis of detailed magnetic susceptibility data.

GEOLOGY AND SAMPLING

The Staroe Slukino section (56.19°N, 42.64°E) is located in a small deep ravine cutting the right bank of the Klyazma River in the village of Slukino (Vladimir Region), 1.25 km upstream of the town of Gorokhovets (Fig. 1). Permian and Triassic boundary rocks were uncovered in three trenches. Trenches Nos. 1540(2167) and 1540(2168) were excavated on the right side of the ravine 190 m below the head of the ravine (the road in Slukino), located 10 m from each other. The composite Permian-Triassic section uncovered there is 13.3 m thick. We studied this section for the first time in 2015. The results of its biostratigraphic analysis were published in (Naumcheva and Golubev, 2019; Naumcheva, 2020; Naumcheva and Golubev, 2020). In 2021, the scheme of the section was updated at the level of beds 14–17, and the upper part of the section with a thickness of 1.65 m—layers 37–41—was studied for the first time. The section exposes the Gorokhovets Member of the Obnora Formation, which is represented by interbedded variegated clays with interlayers of polymictic sandstones and siltstones, less frequently limestones. Remains of numerous ostracods, as well as more rare gastropods, conchostracans, fishes, and plant roots, were found in the rocks. Based on ostracods, the entire sequence belongs to the *Wjatkellina fragiloides—Suchonella typica* zone of the Upper Vyatkian Substage of upper Permian; its lower part (beds 1–29) is related to beds with *Suchonella clivosa* (Nefyodovian Regional Stage), and the upper part (beds 30–41), to beds with *Suchonella rykovi* (Zhukovian Regional Stage) (Naumcheva and Golubev, 2019; Naumcheva, 2020; Naumcheva and Golubev, 2020).

Trench No. 2169 is located on the right side of the ravine, 40 m from trench No. 1540(2167) up the ravine. The Vokhma Formation, represented by interbedded brown polymictic sandstones and red-brown clays and siltstones with plant roots (paleosols), was uncovered there. In the clays, ostracods of the *Darwinula mera—Gerdalia variabilis* zone were found, which indicate that the beds are related to the Vokhmian Regional Stage of the Induan Stage of the Lower Triassic.

Sections No. 1540(2167) and No. 2169 overlap hypsometrically (the overlap is 2.5–2.8 m according to barometric leveling), but do not overlap stratigraphically. Section No. 2169 superimposes section No. 1540(2167) (Fig. 1), but we do not know the size of the stratigraphic hiatus between them because the boundary between the Zhukovian and Vokhmian Regional Stages in Staroe Slukino was not uncovered.

The Staroe Slukino section is structurally similar to the Permian-Triassic red bed terrigenous Zhukov Ravine section located 0.7 km to the south (Golubev et al., 2012; Naumcheva and Golubev, 2019; Naumcheva, 2020; Fetisova et al., 2022). Many of the beds identified in the Zhukov Ravine section are detectable in the Staroe Slukino section. This allows for a detailed lithostratigraphic correlation of the sections, which is confirmed by bio- and magnetostratigraphic data (Fig. 2). However, the Zhukov Member of the Vokhma Formation, which forms the terminal part of the Zhukovian Regional Stage in the Zhukov Ravine section, was not found (not uncovered or missing) in Staroe Slukino. In the Staroe Slukino section, the Zhukovian Regional Stage is represented exclusively by argillaceous rocks. It lacks sandy deposits of channel alluvial genesis, which are present in the Zhukovian Regional Stage in Zhukov Ravine, and, consequently, it lacks indisputable intraformational scour marks and associated local stratigraphic hiatuses. This suggests that the Zhukovian Regional Stage in the Staroe Slukino section is stratigraphically more complete than that in the Zhukov Ravine section.

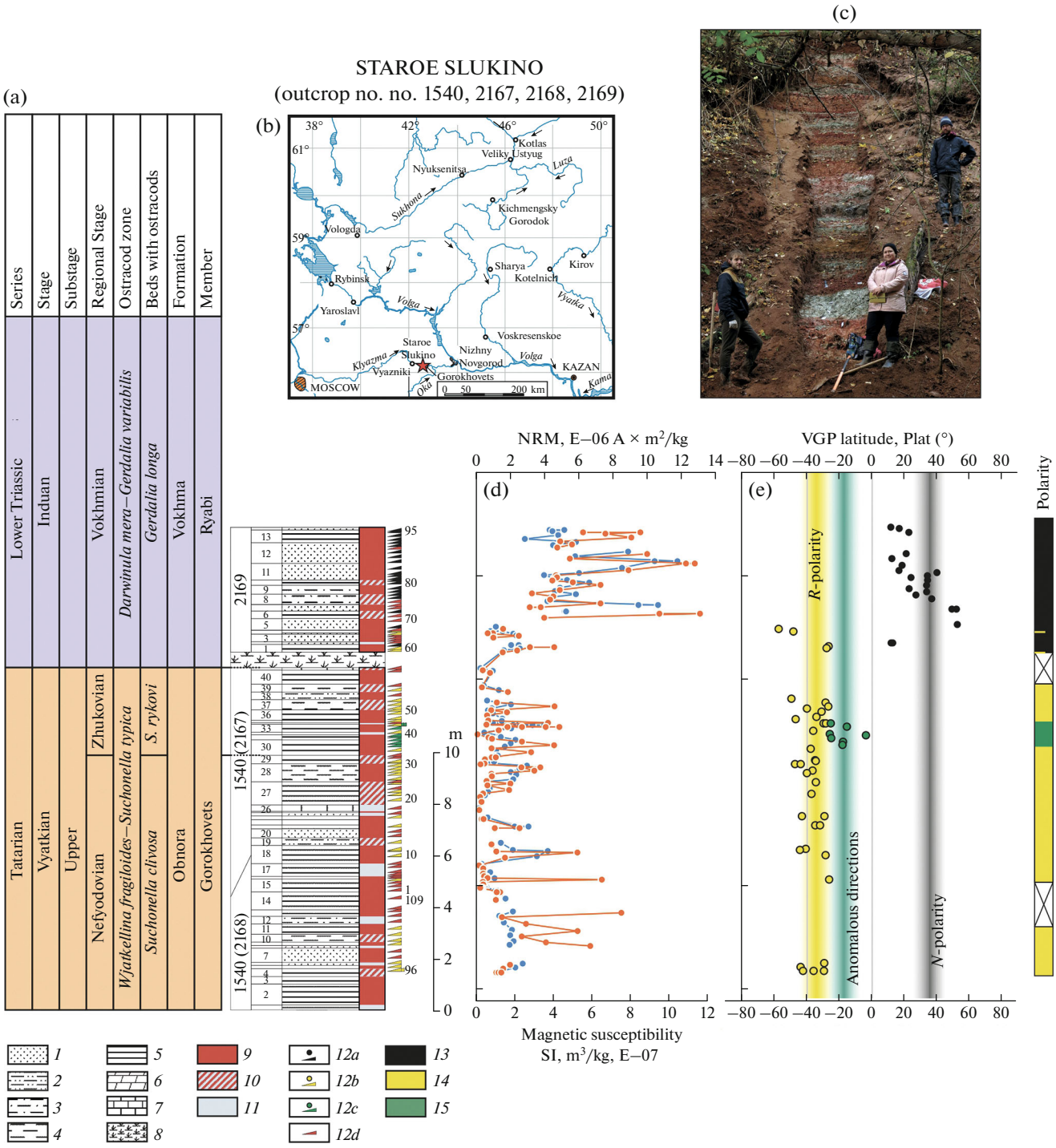


Fig. 1. The Staroe Slukino section: (a) stratigraphy, lithology, and sampling scheme; (b) location on the map of Eastern Europe; (c) photograph of outcrop No. 1540(2167), beds 17–41, the Gorokhovets Member of the Obnora Formation, the Nefyodovian and Zhukovian Regional Stages, upper Vyatkian Substage, the Upper Permian; (d) values of natural remanent magnetization (NRM) and specific magnetic susceptibility; (e) latitude of the virtual geomagnetic pole (VGP). Legend: *1*–*15*—sandstone; *2*—sandy siltstone; *3*—siltstone; *4*—argillaceous siltstone; *5*—clay; *6*—marl; *7*—limestone; *8*—talus; *9*—red beds; *10*—variegated beds; *11*—gray beds; *12*—paleomagnetic sampling levels with a characteristic NRM component: normal polarity (*a*), reversed polarity (*b*), with an anomalous magnetization component (*c*), with noisy paleomagnetic record (*d*); *13*—normal (N) polarity zones; *14*—reversed (R) polarity zones; *15*—zone of anomalous paleomagnetic directions.

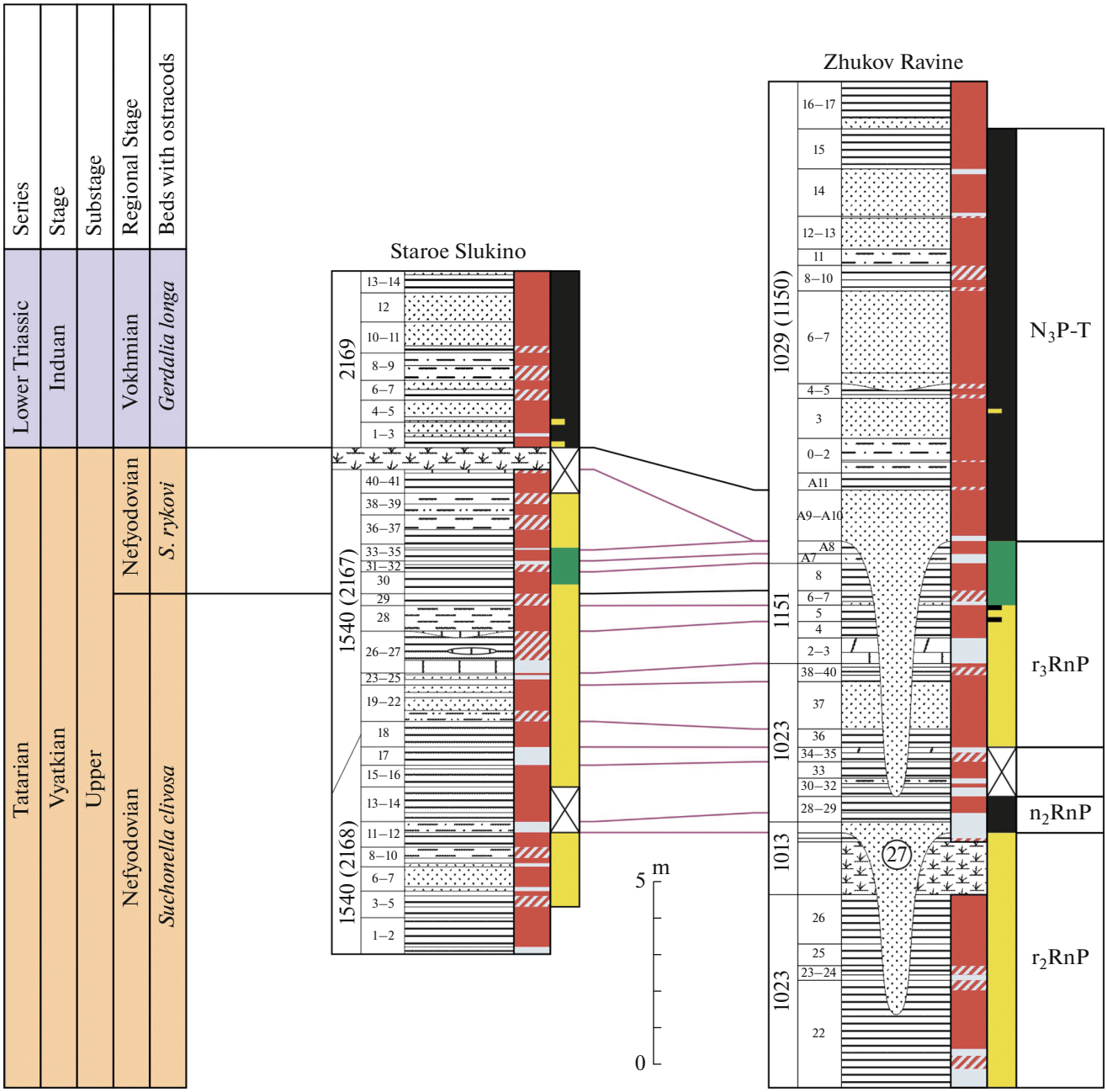


Fig. 2. A correlation diagram for the Staroe Slukino and Zhukov Ravine sections (see Fig. 1 for the legend).

ROCK MAGNETISM AND PALEOMAGNETISM

Analytical Techniques

Magnetostratigraphic studies were carried out with precise layer-by-layer referencing of each oriented sample to the section (Fig. 1). The section was sampled in three trenches, with a total sampling thickness of 16 m; a total of 111 oriented samples were taken, and 113 specimens were subjected to laboratory treatment. The sampling step was from 2 to 20 cm. The samples were taken manually with a knife from trenches 1 m wide and up to 0.5 m deep in the ravine slope; they were oriented by magnetic compass; the value of local magnetic declination was calculated according to the IGRF model (13th generation). One or two oriented paleomagnetic specimens shaped as cubes with a 2 cm edge were cut from each oriented sample. All laboratory studies were carried out according to the standard technique (Tauxe et al., 2018) in the laboratory of the Main Geomagnetic Field and Petromagnetism of the Schmidt Institute of Physics of the Earth of the Russian Academy of Sciences (IPE RAS), the shared

research facilities of the IPE RAS (Veselovskiy et al., 2022), and the Laboratory for Rock Magnetism of the Lomonosov Moscow State University. Anisotropy of magnetic susceptibility (AMS) measurements were made using a KLY-4S Kappabridge (AGICO, Czech Republic) in the field of 200 A/m; the results were interpreted using Anisoft 4.2 software. The dependence of magnetic susceptibility on temperature was measured with a KLY-4S Kappabridge using a CS-4 furnace, and the results were processed using Cureval8 software.

All samples were subjected to detailed progressive thermal demagnetization, which in most cases was performed up to 680°C. There were at least 12 heatings, the number of steps was increased if necessary. The samples were demagnetized using a MMTD-80 non-magnetic furnace (Magnetic Measurements, the United Kingdom) with an uncompensated field value of 5–10 nT. Remanent magnetization was measured using a JR-6 spinner magnetometer (AGICO, Czech Republic). The paleomagnetic analysis of remanent magnetization was processed using the Enkin software package (Enkin, 1994) and PMTools (I.V. Efremov), which use the PCA method (Kirschvink, 1980) to isolate magnetization components. The magnetic susceptibility was measured throughout the section at 5 cm increments with a portable PIMV kappometer (Geodevice LLC, Russia) with a sensitivity of up to 1×10^{-7} SI units.

Rock Magnetism

Scalar magnetic parameters were calculated for all the studied specimens. Specific (mass) natural remanent magnetization in the studied rocks has values in the range from 8.19E–08 to 1.31E–05 A m^2/kg with an average of 2.99E–06 A m^2/kg. The minimum value of specific magnetic susceptibility is 1.58E–07 m^3/kg (SI units), the maximum value is 1.04E–06 m^3/kg (SI units), the average value is 2.29E-07 m^3/kg (SI units). The graph of the specific natural remanent magnetization and specific magnetic susceptibility value distribution over the section thickness clearly shows (Fig. 1d) that there is a sharp increase in the values at the Permian-Triassic boundary: the scalar parameters in the Ryabi Member rocks are significantly higher than those in the Gorokhovets Member rocks. The Koenigsberger ratio (Q) averages 0.26 for the section, with a minimum value of 0.04 and a maximum value of 1.5. The Q ratio is mostly less than 1.0, which is generally expected for sedimentary rocks. A similar behavior of the scalar magnetic parameters was observed in the Zhukov Ravine, Slukino, and Okskiy Siyezd sections and is associated with the increased clastic material flow from the Urals, which experienced an uplift at the beginning of the Triassic (The Permian-Triassic Boundary..., 1998).

Anisotropy of magnetic susceptibility was measured for all specimens to determine magnetic fabric and sedimentation conditions and to have additional control over specimen orientation during sampling and preparation. The degree of anisotropy does not exceed 11%, averaging 4%. The shape of the AMS ellipsoid, except for four samples, is oblate (Fig. 3). The minimum axes (K3) are fairly clustered in the vertical plane, and the maximum (K1) and intermediate (K2) ones are evenly distributed in the lamination plane, which is typical of sedimentary rocks formed under calm hydrodynamic conditions. It should be noted that the distribution pattern of the main axes of the AMS ellipsoid in the samples does not change over the thickness of the section and does not depend on the rock composition, and the uniform distribution indicates the primary nature of the magnetic fabric.

Information on the magnetic mineralogy of rocks was obtained by studying the temperature dependence of the magnetic susceptibility $k(T)$ for the six most representative specimens: there was one heating to 700°C at atmospheric pressure in air, followed by cooling. For four samples, the heating curve is similar to the cooling curve, indicating the absence of significant mineral transformations during heating (Fig. 4, sample 99), and a slight bend, which is observed at 550°C, shows the presence of magnetite in the rocks, possibly with low titanium content. In sample No. 55 (Fig. 4), the heating curve is significantly below the cooling curve. The cooling curve shows a sharp increase in magnetization at 540°C and a dip in the curve at 380°C, which may indicate the formation of sulfides during the experiment. For sample No. 90 (Fig. 4), the heating curve is located significantly higher than the cooling curve, which is characteristic of magnetite transforming into hematite when heated in air.

Paleomagnetism

Two magnetization components are mainly found in the natural remanent magnetization (NRM) of the rocks, and the quality of the paleomagnetic record varies greatly from specimen to specimen. For 45% of the specimens, we can confidently isolate the characteristic magnetization component (ChRM) and calculate its direction, while the paleomagnetic record of 55% of the specimens allows us to determine only the polarity of the ChRM, but its direction is calculated with a MAD of more than 8°; such specimens were not used to calculate the mean paleomagnetic direction. The low-temperature component of the NRM has unblocking temperatures up to 180°C and is most likely viscous in nature (Figs. 5a–5c). The characteristic component isolates in the range of 400–680°C and has reversed (*R*) polarity in the Gorokhovets Member rocks (Fig. 5b), and normal (*N*) polarity in the Ryabi Member rocks (Fig. 5a). The Gorokhovets

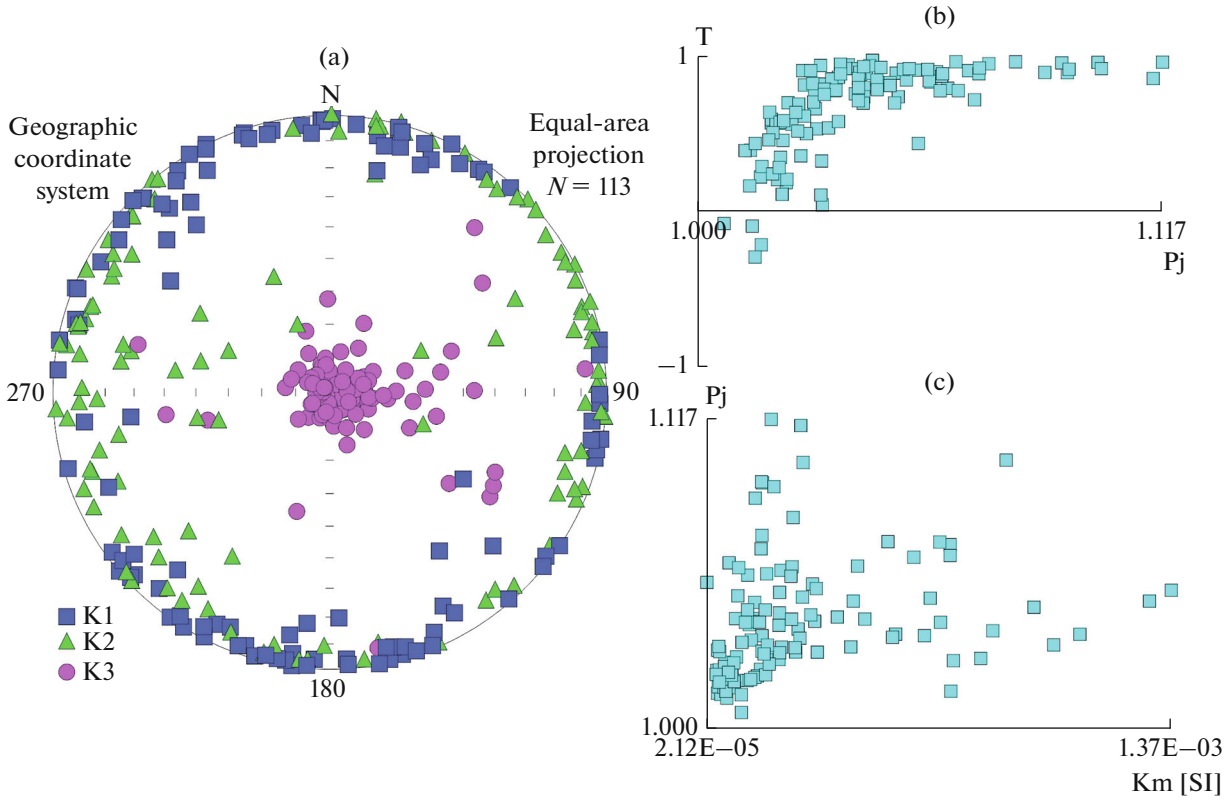


Fig. 3. Anisotropy of magnetic susceptibility (AMS) for rocks of the Staroe Slukino section: (a) a stereogram with the distribution of the AMS ellipsoid axes (K1—maximum axis, K2—intermediate axis, K3—minimum axis); (b) dependence of the shape parameter (T) on the degree of anisotropy (Pj); (c) dependence of magnetic susceptibility values (Km) on the degree of anisotropy (Pj).

Member rocks contain specimens with an anomalous ChRM direction, which are characterized by a low inclination (about 0°) and a declination in the range of 210°–270° (Fig. 5c). The distribution of the ChRM directions in the specimens is characterized by a large scatter, so a cutoff 45° criterion was used to calculate the mean paleomagnetic directions of the normal and reversed polarities (Fig. 5d). The mean paleomagnetic direction calculated with the most stable magnetization components of both polarities is N = 49, D = 51.9°, I=34.2°, K = 24.5, a95 = 4.2° (Fig. 5d). The reversal test (McFadden and McElhinny, 1990) is negative (γ/γ_{cr} = 10.3°/8.6°) for the mean ChRM directions of normal and reversed polarity. The arguments in favor of the primary nature of the isolated components are: 1) there are components of normal and reversed polarity in the rocks of the same section; 2) the calculated mean paleomagnetic direction is close to the expected Permian-Triassic direction (Fetisova et al., 2018b). The paleomagnetic pole corresponding to the calculated mean paleomagnetic direction has the coordinates plat = 36.3°, plong = 155.0°, dp/dm = 2.8°/4.8°, the paleolatitude of the Staroe Slukino section is 18.8° N. The inclination shallowing factor was not assessed due to the small sample size of the single paleomagnetic ChRM directions.

MAGNETOSTRATIGRAPHY

Using the single directions of the characteristic magnetization component in the studied samples, we calculated the coordinates of the virtual geomagnetic poles (VGPs). The magnetic polarity intervals were identified in the section based on the VGP latitude *Plat* (Fig. 1e) and only when the magnetozone was confirmed by data from at least two specimens.

In the upper part of the section, the N_3P-T magnetozone is clearly distinguished by both bio- and magnetostratigraphic data. The lower part of the section, which corresponds to the Gorokhovets Member, correlates with the r_2RnP and r_3RnP reversed magnetic polarity subzones (Fig. 1, 2). In the upper part of the r_3RnP subzone, at the base of the Zhukovian Regional Stage, an interval with an anomalous paleomagnetic record characterized by low values of the VGP latitude was detected. A similar interval was previously identified in several sections of the Russian Platform—Nedubrovo, Zhukov Ravine, and Okskiy Siyezd (Fetisova et al., 2018a, 2022). It must be noted that in the nearby Zhukov Ravine section, this interval is directly overlain by the channel alluvial deposits of the Zhukov Member, which lie erosionally on the bedrock of the Gorokhovets Member. The channel deposits of the

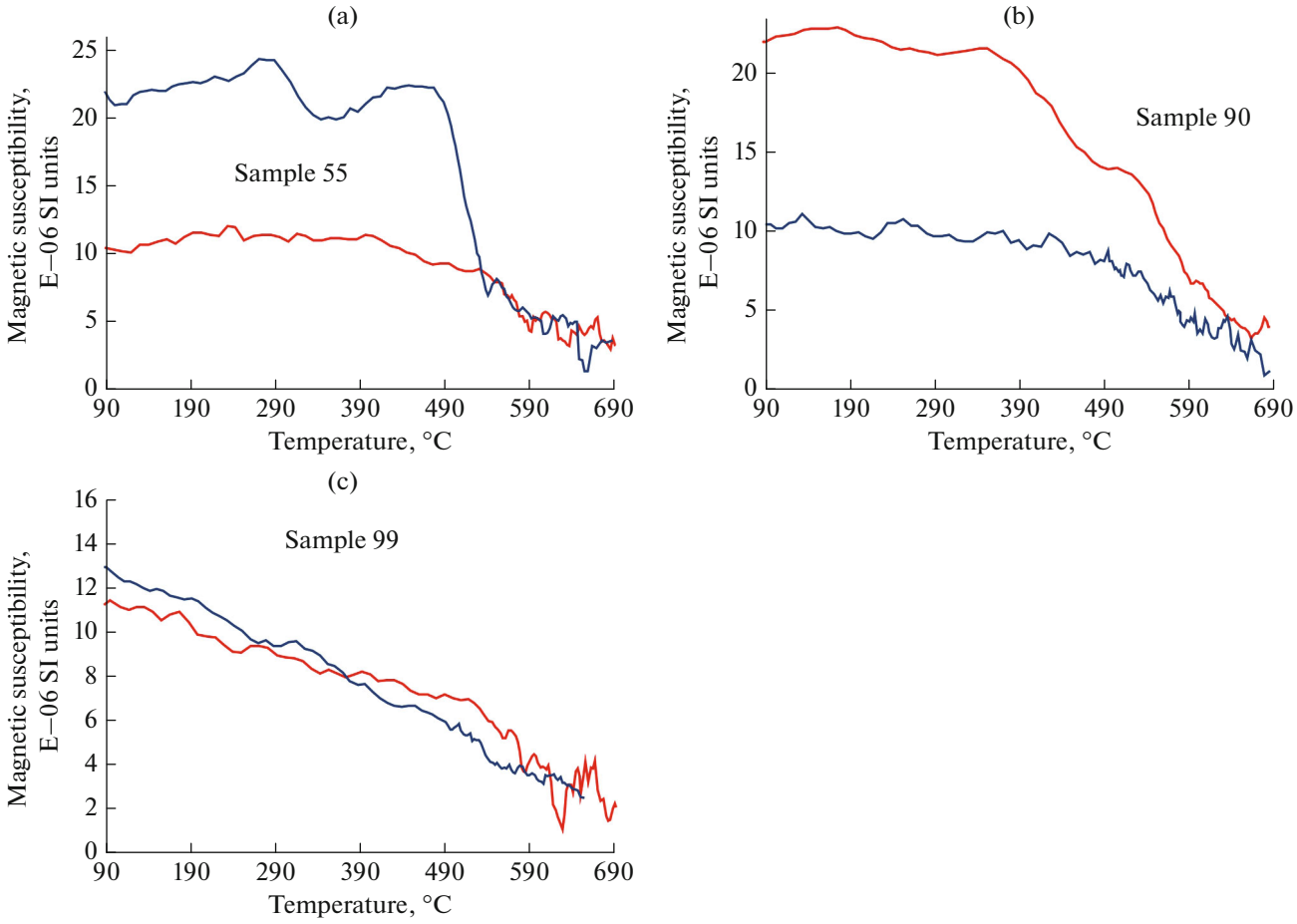


Fig. 4. Temperature dependence of magnetic susceptibility during heating (red curve) and cooling (blue curve) of the sample.

Zhukov Member make the terminal part of the Zhukovian Regional Stage and belong to the N_3P-T orthozone (Fig. 2). Thus, in the Zhukov Ravine section, the interval with paleomagnetic directions is located at the boundary between the intervals of normal and reversed polarity—the r_3RnP and N_3P-T magnetozones. In the Staroe Slukino section, a similar interval with an anomalous paleomagnetic record is located within the r_3RnP subzone. Given the possibility of tracing the beds between the two sections (Fig. 2), we found that, in the Zhukov Ravine section, the deposits corresponding to the reversed polarity zone above the anomalous interval were either insufficiently studied or cut off by the Zhukov Member. Thus, the new data we obtained in this paper specify the stratigraphic position of the anomalous zone (Fig. 6). It should be noted that a similar position of the anomalous paleomagnetic zone within the reversed polarity interval is observed in the Nedubrovo section.

A close examination of Fig. 1d draws attention to the section interval corresponding to the Ryabi Member of the Lower Triassic, for which anomalously low values of VGP latitudes located in the northern hemisphere were obtained. We believe that these data do not indicate an anomalous configuration of the geomagnetic field at the time of rock accumulation because they are not confirmed by the studies of the corresponding intervals in the other sections—Zhukov Ravine, Slukino, and Okskiy Siyezd (Fetisova et al., 2022). It is possible that the shallowed inclinations of the characteristic magnetization component and the low quality of the paleomagnetic record in the red beds of the Ryabi Member are caused by the granulometric composition of the rocks. Indeed, the interval of the low northern VGP latitudes corresponds to a series of sandstones, while the siltstone samples carry the characteristic magnetization component of the expected direction.

A curious case is the position of the anomalous paleomagnetic record interval in the Okskiy Siyezd section. Here we (Fetisova et al., 2022) also found an interval with anomalous paleomagnetic directions inside the r_3RnP reversed polarity interval. However, based on the biostratigraphic data, we concluded that it is located below the anomalous interval of the Zhukov Ravine section—not in the Zhukovian Regional

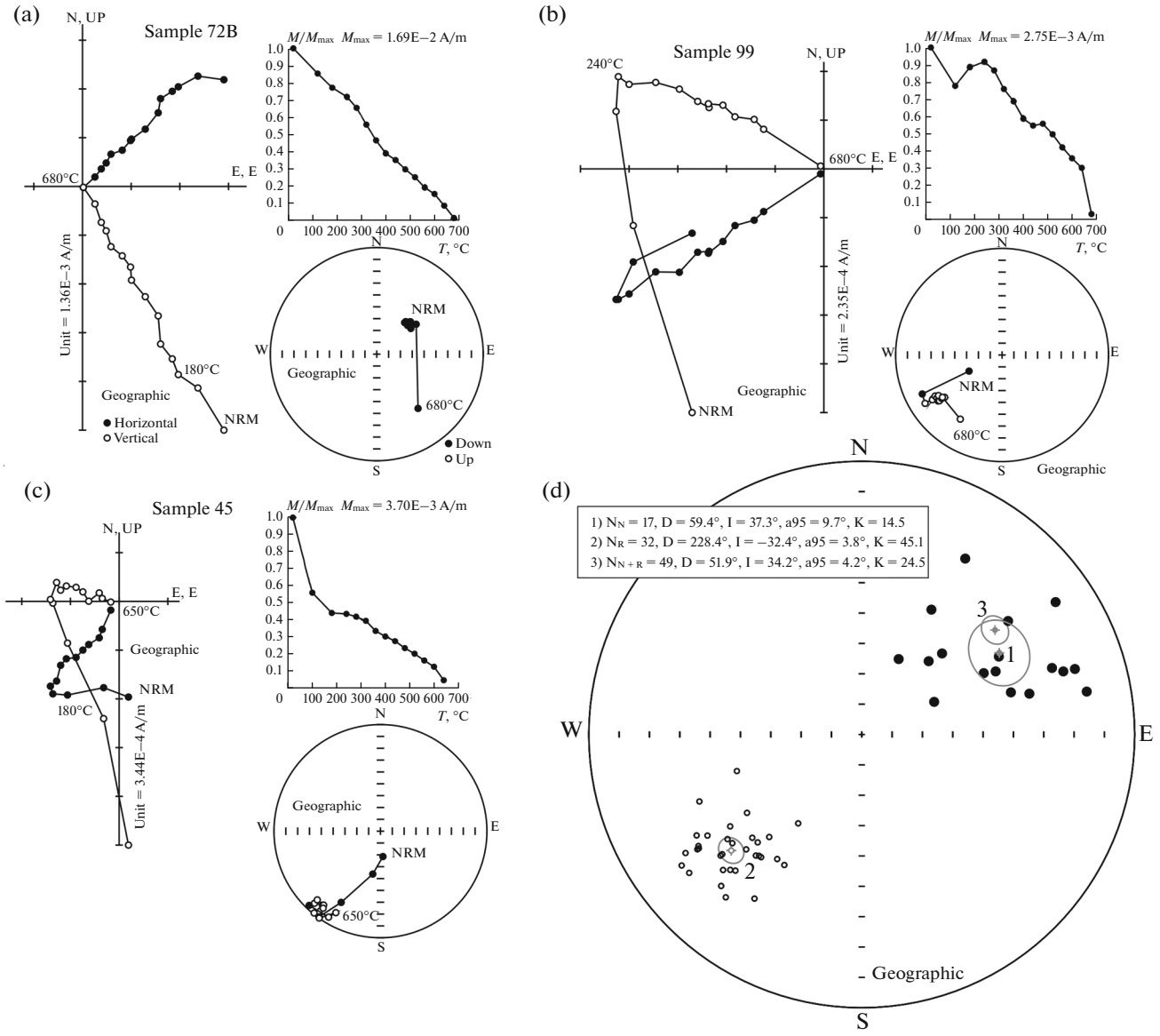


Fig. 5. Stereograms, demagnetization curves and Zijderveld diagrams showing the process of stepwise thermal demagnetization of the samples. The geographic (present-day) coordinate system: (a) an example sample with the ChRM of normal polarity; (b) an example sample with the ChRM of reversed polarity; (c) an example sample with the anomalous direction of the ChRM; (d) the distribution of single directions of the characteristic magnetization component (ChRM) and mean directions of the NRM components of normal (No. 1), reversed (No. 2), and normal and reversed (No. 3) polarities with 95% confidence circles.

Stage, but in the Nefyodovian Regional Stage. Thus, the composite magnetostratigraphic section of the Upper Vyatkian Substage of the Oka River basin contained two zones of anomalous paleomagnetic directions similar in their characteristics (Fetisova et al., 2022). Since the latter is highly unlikely, we reanalyzed the biostratigraphic data for the Okskiy Siyezd section, the results of which are presented below.

Fossil remains of tetrapods (Lagerny Ovrag-2 site) and ostracods were found in the Okskiy Siyezd section, in the r_3RnP magnetozone above the interval with anomalous paleomagnetic directions. These findings indicated the Nefyodovian age of the bone-bearing deposits (Golubev et al., 2012; Naumcheva and Golubev, 2019; Naumcheva, 2020). The Pre-Zhukovian age of the tetrapod site is not questioned, since a tusk of *Inostranzevia* sp. was found here. This genus is characteristic only for the *Scutosaurus karpinskii* tetrapod assemblage zone (Sennikov and Golubev, 2017), the upper boundary of which coincides with the boundary of the Nefyodovian and Zhukovian Regional Stages (Boyarinova et al., 2022). Remains of tetrapods were found in a sandstone bed with an exposed thickness of over 2 m, which was tem-

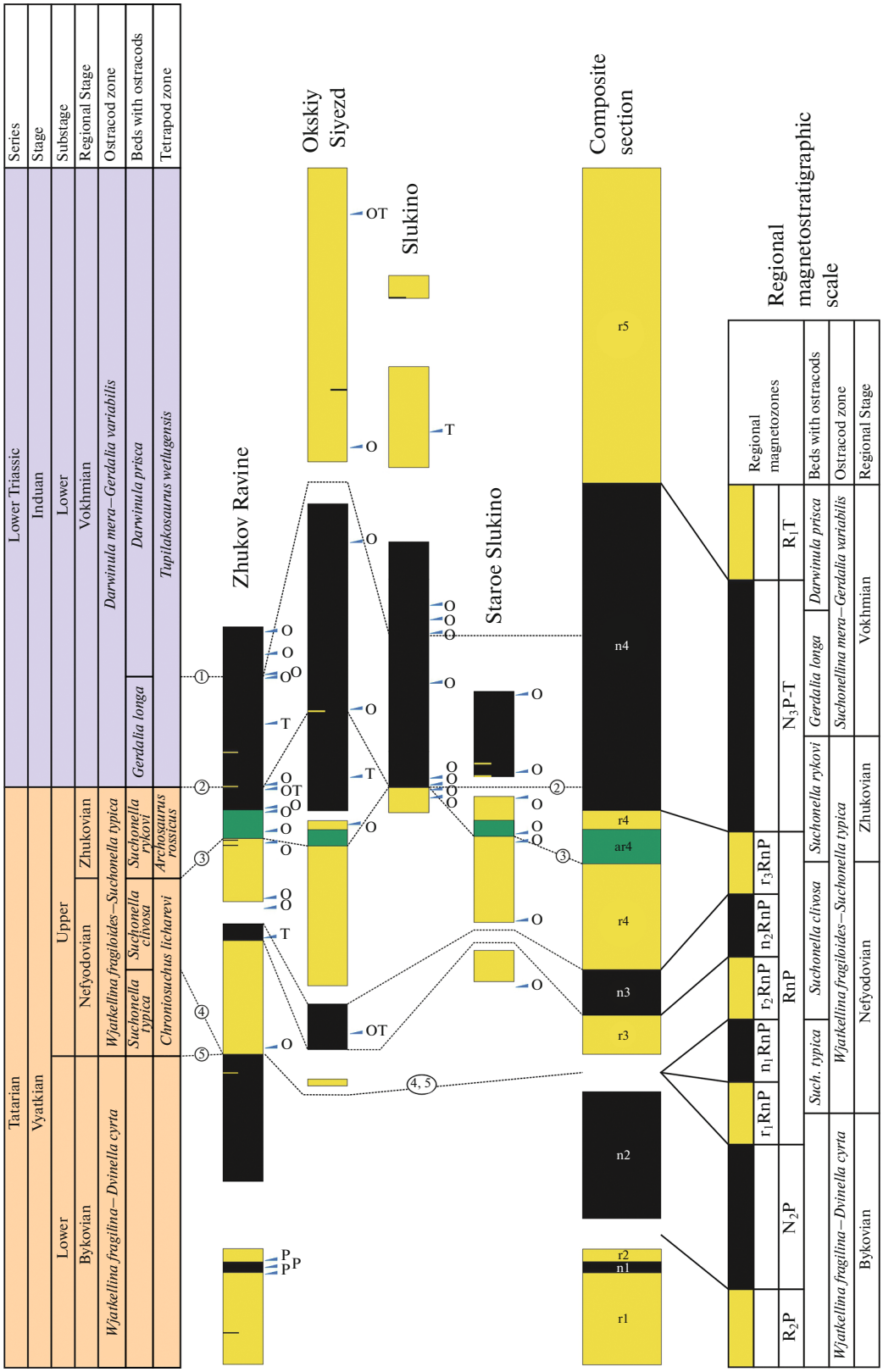


Fig. 6. The correlation of the Permian-Triassic boundary magnetostratigraphic sections of the Oka River basin, the composite magnetostratigraphic section, and the regional magnetostratigraphic scale with indices of magnetozones. Levels of findings of stratigraphically significant fossil remains: O—ostracods, T—tetrapods, P—fish. Correlation levels—lower boundaries of: 1—beds with *Darwinula prisca*; 2—beds with *Gerdalia longa* (Vokhmian Regional Stage); 3—beds with *Suchonella rykovi* (Zhukovian Regional Stage); 4—beds with *Suchonella clivosa*; 5—beds with *Suchonella typica* (Nefyodovian Regional Stage and Upper Vyatkian Substage). See Fig. 1. for the legend.

porarily exposed during road construction in 2011 on the left slope of the northern branch valley of the Lagerny ravine, section No. 1102 (the right bank of the Oka River in Nizhny Novgorod). Paleomagnetic specimens were taken from a section No. 1135J located on the right side of the same branch valley, 60 m west-northwest of the Lagerny Ovrag-2 site. A lack of continuous bedrock outcrop between these sections prevented their lithostratigraphic correlation during field studies. The position of the bone-bearing sandstone in the section No. 1135J was then determined by barometric leveling with a barometer built into the GPS receiver Garmin GPSmap 76CSx. At the hypsometric level corresponding to the level of the bone-bearing sandstone, there are four sand beds from 0.1 to 0.3 m thick here in the 2.5 m interval. One of these beds (bed 11) is located below the interval with anomalous paleomagnetic directions, one bed (bed 9)—inside the interval, and two beds (beds 5 and 7)—above it. The uppermost of these sand beds was found to be analogous to the bone-bearing sandstone. However, given the low accuracy of the barometric leveling performed and the possible non-horizontal bedding of the rocks, any of these beds may be analogous to the bone-bearing sandstone. Moreover, the bone-bearing sandstone can wedge out, and then there is nothing analogous to it in the section No. 1135J. Thus, the stratigraphic position of the Lagerny Ovrag-2 tetrapod site relative to the beds with the anomalous paleomagnetic record should be recognized as uncertain.

Ostracods were found in the same outcrop in which the paleomagnetic samples were taken. The clay with ostracods (bed 1105/9 (Naumcheva and Golubev, 2019; Naumcheva, 2020)) is located 1.3 m above the anomalous interval, in the stratum of reversed polarity rocks. The assemblage of ostracods includes *Volganella truncata, Suchonella circula, S.* aff. *auriculata, Wjatkellina verbitskajae*, and *Suchonellina compacta* (Naumcheva, 2020). With the exception of *S.* aff. *auriculata*, all the taxa are characteristic of both the Nefyodovian (beds with *Suchonella clivosa*) and the Zhukovian (beds with *Suchonella rykovi*) Regional Stages (Naumcheva and Golubev, 2019; Naumcheva, 2020). *S.* aff. *auriculata* in the Moscow Syneclise was found in three sections: Aristovo (Vologda Region), Slukino and Staroe Slukino. In these sections, it is indeed common only in the Nefyodovian Regional Stage (Naumcheva, 2020). However, the available stratigraphic data on this unusual ostracod species are clearly insufficient to confidently diagnose the Nefyodovian Regional Stage based on the presence of this taxon alone.

Thus, the current biostratigraphic data do not allow us to determine the exact age of the beds with anomalous paleomagnetic directions in the Okskiy Siyezd section. These beds may correspond to the Nefyodovian Regional Stage or the Zhukovian Regional Stage alike. Based on the foregoing, we recognize the presence of only one anomalous interval ar4 in the composite magnetostratigraphic section of the Upper Vyatkian of the Oka River basin (Fig. 6). This interval is located within the r_3RnP subzone at the base of the Zhukovian Regional Stage.

CYCLOSTRATIGRAPHY

Technique

The cyclostratigraphic approach to estimating how long it takes sedimentary rock sections to accumulate is based on climate cyclical nature as a consequence of changes in the Earth's orbit parameters of known duration, i.e., Milankovitch cycles. The value of magnetic susceptibility (MS), which is dependent on climatic conditions during sedimentation, makes it possible to distinguish Milankovitch cycles by means of spectral analysis methods. The number of identified cycles of different order determines how long it took the sedimentary strata to form.

Magnetic susceptibility measurements for cyclostratigraphic studies were carried out on the natural outcrop with a portable kappometer at ~5 cm increments, the kappometer's positioning error is estimated to be ±1 cm. It should be noted that magnetic susceptibility was also measured in the laboratory for all the specimens taken for paleomagnetic studies, and a comparison of its values obtained by two different measurement methods showed an excellent repeatability of the results.

The principles of the approach we used are described in detail in the paper by (Kodama and Hinnov, 2014), and we strictly follow all of its steps. The first step in the cyclostratigraphic analysis of the Staroe Slukino section is to assess the effect of the measurement point positioning error on the final result. For this purpose, we generated 10 random sets of intervals of magnetic susceptibility measurements based on its normal distribution with the following parameters: mean value—5, standard deviation—1 (Fig. 7a). For each resulting curve, MS measurements were virtually taken at equal 5 cm increments; after that a long-period trend built using the LOESS (local polynomial regression) method with a window equal to 66% of the sequence length was subtracted from the curve. This procedure makes it possible to remove the peaks corresponding to possible period lengths repeated less than 2 times in the signal from the spectrogram. After subtracting the trend, each curve was subjected to a spectral analysis procedure according to the MTM method (Thomson, 1982), which provides an averaged estimate of spectral density by applying a set of weight functions represented by orthogonal spheroidal functions to the time series (Slepian and Pollak, 1961). To assess the reliability of the selected periods, we tested the null hypothesis that the entire signal is red noise. Red noise is understood to be a ran-

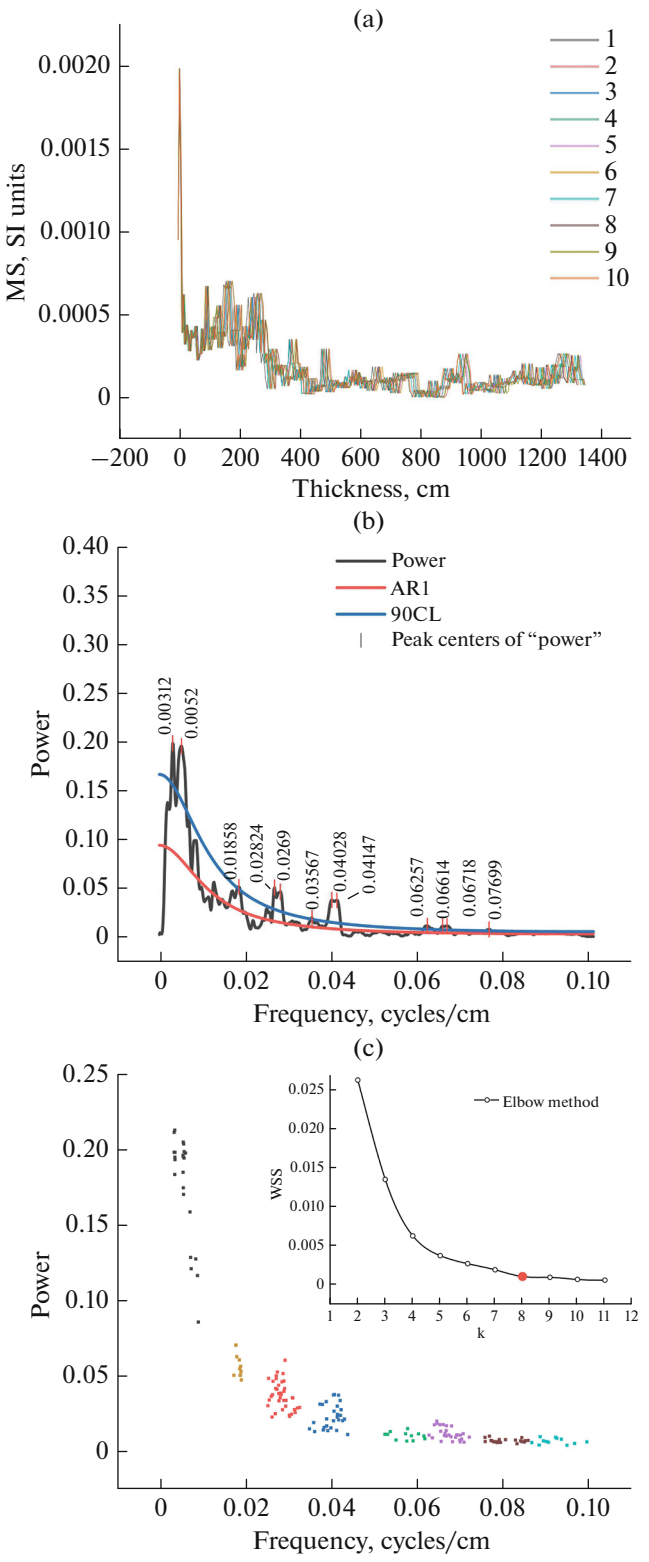


Fig. 7. The results of cyclostratigraphic studies: (a) 10 generations of sampling with increments from the normal distribution with the parameters mean = 5, standard deviation = 1; (b) an example periodogram for MS variations with a red noise level and a confidence level of 90%; (c) clusters of peaks (different colors) in the periodograms of MS variations.

dom process that has some "memory"—each observation in it has a positive correlation with the previous one. The power spectrum of such a process has an increased density in the low frequency range, which is reflected in the name of the noise by the word "red". These are the processes that describe many climatic fluctuations on large time scales (Hasselman, 1976). The periodograms contain the mean red noise spectrum and the confidence levels (CLs) corresponding to various probabilities of difference from the null hypothesis. For all the periodograms, the peaks above 90% (Fig. 7b) were combined into clusters by the *K*-means method; the *K* value was determined by the "elbow" method and is equal to 8 (Fig. 7c). The spectral analysis was performed using Acycle v.2.4 software (Li et al., 2019).

RESULTS AND DISCUSSION

The mean values of period lengths for the identified clusters of peak frequencies in the periodograms are as follows: (1) 187.8 cm; (2) 54.9 cm; (3) 35.4 cm; (4) 25.1 cm; (5) 17.5 cm; (6) 14.9 cm; (7) 12.5 cm; (8) 10.9 cm (Table 1). In this case, the clusters for frequencies above 0.05 (clusters 4–8) cannot be clearly distinguished: for different variants of the magnetic susceptibility curves, the positions of the peaks differ greatly, indicating that they are very prone to the measurement positioning error in the section, so they were excluded from consideration. The first four clusters correspond to periods with length ratios of 1.00/3.43/5.31/7.48 (Table 1). According to the La04 model (Laskar et al., 2004), the duration of Milankovitch cycles for an age of 250 Ma is 125.0 thousand years; 95.0 thousand years (eccentricity), 36.3 thousand years (tilt of the Earth's rotation axis to the ecliptic), 22.0 thousand years; 20.9 thousand years; 17.9 thousand years (precession), and the ratios of their lengths are close to those for the periods that we identified. Therefore, we can clearly correlate cluster 1 to cycles of eccentricity, cluster 2 to cycles of tilt of the Earth's rotation axis, and clusters 3 and 4 to cycles of precession. This allows us to calculate that the sampled interval of the section took 900 ± 20 thousand years to accumulate.

It must be noted that, since the thickness of the interval with anomalous paleomagnetic directions at the base of the Zhukovian Regional Stage is about 2 m, the above calculation of how long it took the entire 16 m section to accumulate allows us to estimate that the geomagnetic field was in an anomalous state in the Late Permian for about 110 thousand years, which agrees with an independent estimate of this interval's duration in the Nedubrovo section (Fetisova et al., 2018).

Table 1. A correlation of the identified periods to Milankovitch cycles

Periods in magnetic susceptibility variations					La04 (Laskar et al., 2004)	
No.	frequency	period, cm	ratio	calculated duration, thousand years	ratio of Milankovitch cycles	duration of Milankovitch cycles, thousand years
1	0.0053	187.97	1.00	124.00	1.00	125
2	0.0182	54.85	3.43	36.19	3.44	36.3
3	0.0282	35.41	5.31	23.36	5.68	22
					5.98	20.9
4	0.0398	25.11	7.48	16.57	6.98	17.9
5	0.0572	17.49	10.74	11.54		
6	0.0672	14.87	12.64	9.81		
7	0.0803	12.46	15.09	8.22		
8	0.0918	10.90	17.25	7.19		

CONCLUSIONS

The studies allowed us for the first time to obtain rock magnetic and paleomagnetic properties of the Permian and Triassic boundary deposits of the Staroe Slukino section, on the basis of which a magnetic polarity scale was made for this section and correlated with the Permian and Triassic sections of the Russian Platform's central regions and the regional magnetostratigraphic scale. The main particular results of the study are as follows:

1) The structure and position of the r_3RnP reversed polarity subzone are specified. Within the subzone at the base of the Zhukovian Regional Stage, there is one interval of anomalous paleomagnetic directions, above which the rocks of the middle part of the Zhukovian Regional Stage are magnetized in the reversed polarity.

2) It is estimated by the cyclostratigraphic method that the sampled 16-meter part of the Staroe Slukino section took 900 ± 20 thousand years to accumulate, and the geomagnetic field anomalous state interval lasted for 110 thousand years.

3) A new Permian-Triassic paleomagnetic pole of the East European Platform is calculated.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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